ETHAN C. GOLDSTEIN

Rockville, MD | Ethancgoldstein@gmail.com | 301.526.0790
ethangoldstein.dev | github.com/ethan-goldstein | linkedin.com/in/ethangoldstein-

Computer Information Systems student and sole software engineer at SafeRides, a campus rideshare startup, owning the rider app, driver app, backend, and admin panel. Also builds across systems programming, applied machine learning, and signal processing. Public Trust clearance adjudicated 2026 (U.S. Treasury/IRS), with VA reciprocity. Validates every result by trying to break it first.

EDUCATION

University of South Carolina *Columbia, SC*
B.S. Computer Information Systems, Minor in Business Information Management *Expected May 2027*
Coursework: Data Structures and Algorithms (Java), Information Security Principles, Web Applications, Computer Hardware Foundations, Capstone Computing Project I, Professional Issues in CS & Engineering
Study Abroad: Florence University of the Arts, Firenze, Italy *Spring 2026*

EXPERIENCE

SafeRides *2026 to Present*
Software Engineer · Sole engineer, reporting to the Founder & CEO · Coetzee Tech Inc. *Remote*
- Only engineer on a campus rideshare platform, owning the rider app, driver app, backend services, and admin panel end to end.
- Own production: diagnose and fix bugs, monitor and resolve live errors, and run releases and deployments with pre-release testing.
- Restructure the codebase for whoever comes next: module boundaries, naming, dead-code removal, behavior-preserving refactors, clean version control, and documentation so a future engineer can continue the work.

AI & Workflow Automation Engineer *2025 to Present*
Independent Practice · github.com/ethan-goldstein/Autonomous-OS *Remote*
- Operate an 11-agent automation platform unattended on self-hosted hardware, architected so every agent inherits cron scheduling, node:sqlite persistence, and live log streaming from a shared base class, making a new agent one file and a registry line.
- Built its two-tier model dispatcher, Claude Code CLI primary with local Ollama fallback behind a 10-minute persisted circuit breaker, and capped inference at 2 concurrent processes on a FIFO semaphore after unbounded forking crashed an 8GB machine.

GovCIO *2026*
Data Processor · Veteran & Enterprise Technology Sector *Tysons Corner, VA*
- Held a Public Trust clearance (U.S. Treasury, IRS) extended to Department of Veterans Affairs systems by background-investigation reciprocity, PIV-sponsored through GSA USAccess.
- Processed and validated confidential Veteran records under the VA HTMS contract in secured remote environments.
- Ran high-volume digitization workflows across multiple IRS document types: scanning, indexing, and quality control.

Oxford Government Consulting *2025*
Data Entry Operator *Tysons Corner, VA*
- Supported the IRS Digitalization-as-a-Service modernization program in partnership with GovCIO, on a pipeline processing 2 million document images per day, and flagged processing-software inefficiencies for upgrade.

The Baseball Zone, *Manager*. Maintained HitTrax hardware and software on site; ran front desk operations, scheduling, and daily finances. *2021 to 2024*

PROJECTS

SENTINEL: Gesture-Controlled Follow-Me Drone *Python, MediaPipe, OpenCV, Kalman filter, COLMAP*
- Hands-free gesture flight for a $60 Wi-Fi drone behind a safety governor. Its person-follow lock held on 84% of real flight footage, up from 8%, via pose tracking and a Kalman filter. 202 tests.

PARALLAX: Bitemporal Analytical Engine *C++20, WebAssembly, SIMD, libFuzzer*
- Query language written end to end, lexer through cost-based planner, over a columnar bitemporal store with a Z-order index. Policy refusal at plan time, sound because two of four cardinality estimators are exact. Differential tested against a naive oracle.

ORRERY: Atlas of Worlds from Real Data *TypeScript, three.js, Web Workers, Playwright*
- Eight worlds from real data: planets and 22 moons tested against JPL Horizons, 19,799 satellites on SGP4 in a worker, 100k GPU-advected ocean-current particles. 121 unit tests plus Playwright and axe in CI.

NULLHYP: Quantitative Research Engine *TypeScript, Web Workers, zero-dependency ML, Python cross-check*
- Reports no edge: Sharpe 2.46 in sample, -0.18 out, deflated Sharpe 0.06 over 12 trials. Validation came before any model: purged K-fold with embargo, combinatorial purged cross-validation, and a lookahead audit that recomputes every feature on data truncated at bar t. Trees, elastic net, and logistic regression from scratch in a Web Worker, matched on 22 of 22 checks by a pandas build.

SKILLS

Languages & Verification: C++20, Java, Python, TypeScript, SQL, Bash; libFuzzer, ASan and UBSan, differential testing
AI, ML & Statistics: multi-agent orchestration, local LLM inference (Ollama), Claude Code CLI, prompt-injection mitigation; gradient boosting, elastic net and logistic regression from scratch, purged and combinatorial cross-validation, permutation tests, DSP
Backend & Distributed Systems: Node.js, Express, WebSockets, Server-Sent Events, node:sqlite (WAL), REST, OAuth 2.0, HMAC-SHA256 auth, Cloudflare Workers and Durable Objects, Tailscale, launchd and cron, Playwright, Git, GitHub Actions CI/CD
Frontend: React, Next.js App Router, Vite, Three.js, WebGL, WebAssembly, GSAP, MediaPipe, code splitting, WCAG accessibility
Clearance: Public Trust, U.S. Treasury (IRS), adjudicated 2026; VA reciprocity, Veteran and Enterprise Technology Sector